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PALAEOMAGNETIC PROPERTIES OF THE MIDDLE-UPPER DEVONIAN VOLCANICS OF THE ORKNEY ISLANDS*

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Thermal demagnetization and ore microscopy combined with the magnetic colloid technique and measurements of the temperature dependence of saturation magnetization have been carried out on the Middle-Upper Devonian Volcanics of the Orkney Islands. The investigations show that the original remanence has been replaced by subsequent chemical magnetizations. These low temperature processes probably encompassed a long time interval, ranging upwards to about 50 my. Switching of the Earth's field polarity during this time yielded composite magnetizations resulting in intermediate or deviating natural remanence directions. Reliable estimates of separate components may pose severe experimental problems. However, some few acceptable data have been extracted tentatively suggesting an ancient relative pole position at 149.5° E, 23.7° N. This result is in good agreement with polar estimates based on the Norwegian Old Red Sandstone rocks. Furthermore, there is evidence that the Orkney volcanics originated in a reversed polarity epoch. This latter suggestion fits into a Devonian polarity/time sequence as previously suggested.

1. Introduction

Interpretation of the palaeomagnetic data of the Devonian of Europe has been a subject of considerable controversy in recent years. The main problem in the discussion has been whether the Lower Devonian lavas in the Caledonian basin of southern Scotland (with diverging magnetization directions) reflect a true Devonian geomagnetic field [1–4] or whether they constitute false (composite) magnetization directions brought about by remagnetization [5–9]. The latter suggestion is supported by detailed demagnetization studies of a number of different formations including also Lower Devonian lavas of Great Britain (the Cheviot complex). In its ultimate consequence the theory of remagnetization [8] implies that strongly deviating results may develop if the period of remagnetization covers a time span of mixed geomagnetic polarity (the magnetization axis may be constant). Despite the problem of estimating the original polarity in many cases there is some evidence of a single polarity inversion (or a short period of mixed polarities) in the Devonian, the transition period occurring around early Middle Devonian [7]. This trend in polarity as a function of rock age suggests that the stable components deduced (or indicated) are not essentially younger than the original age of the rocks. If this polarity/time sequence is proved by further studies, it may represent a very important tool in Old Red Sandstone correlation.

2. General geology and sampling

The Orkneys are a group of islands separated from the mainland of Scotland by the Pentland Firth. A

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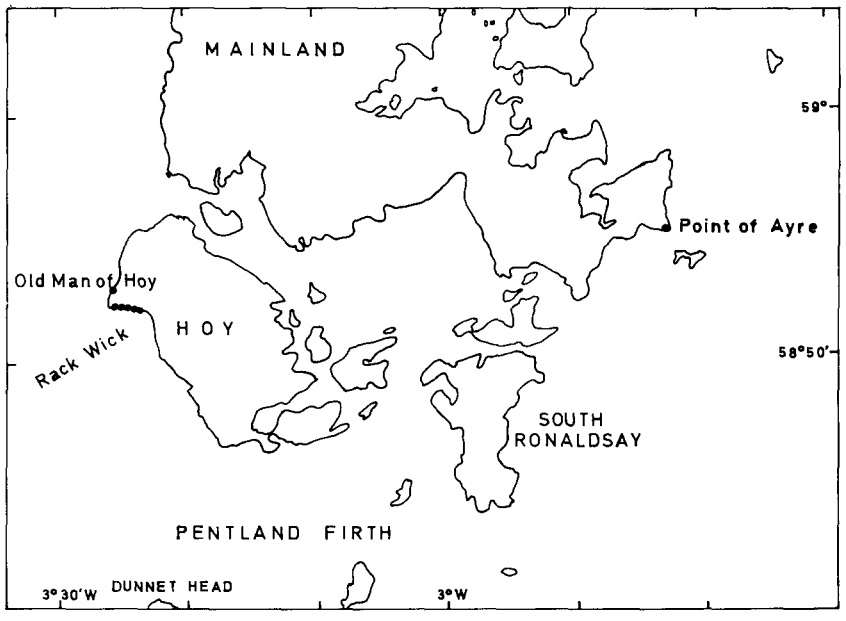


Fig. 1. Sketch map of parts of the Orkney Islands. Sampling localities are marked by dots.

summary of the geology of these islands is given by Wilson et al. [10].

The Orkneys are nearly exclusively built up of flagstones and sandstones belonging to the Middle and Upper divisions of the Old Red Sandstone, the total stratigraphic thickness spanning about 5000 m. The geological structure of the islands is fairly simple. Despite a large number of dissecting faults, the greater number of them trending north-northeast, the attitude of the rocks is mostly horizontal or gently dipping.

A thin volcanic zone consisting of at least two lava flows is found in the upper section of the Middle Old Red beds of East Mainland, Shapinsay and the Black Holm of Copinsay. The lava flows are dark green, highly decomposed basic rocks which are in part vesicular and contain pipe amygdales. There are some indications [10] that these lavas were laid down on a wet surface and became covered with sediments soon after extrusion.

A second volcanic zone occurs at the base of the Upper Old Red Sandstone of Hoy. Previous to this later igneous activity the area of Hoy had been subjected to considerable tectonic activity with faulting, folding and uplift. The end of the period of subsequent erosion was accompanied by an outburst of volcanic activity. This volcanicity was of explosive character and an ashy deposit with a thickness ranging up to about 15 m was laid down before the succeeding lava flow (or flows) was poured out. In the various lava outcrops of Hoy only one flow appears to exist at any one locality and the rock type is always the same: an olivine-basalt with phenocrysts of olivine and felspar.

The attitude of the rocks in the outcrops concerned is very nearly horizontal. The sampling localities are shown in fig. 1. The rock collection comprises altogether 30 hand samples.

3. Magneto-mineralogical considerations

The stability of titanomagnetites, which at least are responsible for the primary magnetization of basaltic rocks, is strongly dependent on the oxygen pressure. Thus, the equilibrium oxygen partial pressure for primary titanomagnetites as surface temperatures appears to be as low as 10^{-60} atm. [11]. This means that titanomagnetites exposed at the Earth's surface have a tendency to be oxidized to the stable phases in air: haematite and rutile.

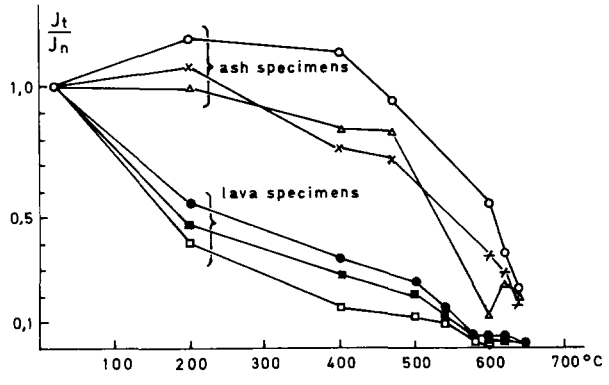


Fig. 2. Examples of the decay of remanence intensity vs. temperature illustrating the magnetic stability difference between the lavas and the ashy sediment.

According to Readman and O'Reilly [12], oxidation of synthetic titanomagnetite readily takes place at 200°C forming titanomaghemite which is further transformed into a mixture of haematite and rutile. It is likely that these oxidation processes also take place at surface temperatures though the reaction rate will be slower. Following the idea of Readman and O'Reilly [12] the process of low temperature oxidation in titanomagnetites exposed to high oxygen partial pressure may schematically be represented as follows:

Low temperature oxidation

titano-magnetite —oxidation→ titano-maghemite (cation deficient spinel)

transformation → haematite + rutile (and/or anatas)

An important feature which appears to develop during low temperature oxidation has been recognised by Ade-Hall et al. [13] in a survey on Icelandic lavas. It was found that the titanomagnetites of lava flows having experienced a burial of 900 m have been transformed into an assemblage of mainly small rutile particles (a so called 'granulation' structure).

Slowly-growing magnetic phases at low temperature may build up a chemical magnetization representing a long time interval. A proper illustration of the magnetic complexity arising in such cases has recently been demonstrated in Permian lava flows of southern Norway [14].

As seen below, the Orkney volcanics provide another striking example of severe low temperature transformations of the iron-titanium oxides. All collected samples from the volcanics have been subjected to ore microscopy. In addition, the colloid technique [15] has been applied in order to distinguish between strongly magnetic and weakly magnetic (or non-magnetic) phases. Furthermore, the temperature dependence of saturation magnetization (in 5000 Oe) has been measured on some specimens. The results are as follows:

(1) *Lava specimens.* In all polished sections studied the iron-titanium oxides have been strongly altered. No trace of the primary titano-magnetite has been recognised. The important low temperature alteration of the opaques is of two different types.

(i) A fine-grained assemblage which appears to consist mainly of rutile (anatas?). The granulated grains do not attract colloid, indicating, therefore, the absence of a spinel phase.

(ii) A fine-grained intergrowth of different minerals such as rutile (anatas?), haematite and a titanium-poor spinel (probably titanomaghemite). Although this intergrowth occurs on such a fine scale that the different phases can not be optically diagnosticated with a high degree of certainty, the presence of haematite and of the spinel phase is clearly recognized by the magnetic measurements (the latter phase having a T_C of about 500°C). The ore grains consisting of this type of mineral assemblage clearly attract the magnetic colloid. The presence of a titanium-poor spinel shows that the stage of equilibrium has not been reached.

Some lava samples show signs of high temperature deuteric oxidation, but these features are always strongly overprinted by the subsequent low temperature processes. Thus, studies of the magnetic mineralogy suggest strongly that the original thermoremanence has been completely replaced by subsequent chemical magnetizations. By considering the gradual process of oxidation (as schematically represented above) it seems possible that the remanence carried

by the spinel component predate that carried by haematite. As shown below, the palaeomagnetic data tend to confirm this suggestion.

(2) *Ashy specimens.* The ore grains of the ashy deposit have reached an ultimate state of oxidation as all grains are completely oxidized to haematite and rutile (ilmohaematite?). No grains attract the magnetic colloid. The originally very loose character of this deposit probably enabled the oxidation to proceed very rapidly. Therefore, a major fraction of haematite was possibly formed fairly soon after deposition.

4. Demagnetization experiments

A total of about 60 specimens from the Orkney volcanics have been subjected to progressive thermal demagnetization. After each heating step the specimens (5–12 in each heating run) were cooled down to room temperature in zero field for measurement with an astatic magnetometer. The lavas were partly heated in air and partly in an inert atmosphere (argon) while the ashy sediment has been heated in air exclusively. The intensity of natural remanent magnetization (NRM) of the two types of rock is quite different, being mostly of the order of 10^{-4} emu/cm^3 for lavas and of the order of 10^{-6} emu/cm^3 for the ashy sediment. The difference in oxidation state of the two rock types concerned has magnetically been recognized in different ways. First of all there is a marked contrast in thermal stability, the remanence intensity of the lava specimens being commonly reduced by about 50 per cent after heating to 200°C while the ashy sediment had to be thermally demagnetized to about 600°C to achieve a similar intensity reduction. Examples of this difference in the decay of remanence intensity as a function of temperature are shown in fig. 2. Furthermore, there appears to be no doubt that the dominant remanence-carrying mineral in the volcanics ashes is haematite, while a spinel phase contributes the essential remanence intensity of the lavas. This observation agrees completely with the ore microscopy data.

As suggested by the ore mineral considerations the magnetization possessed by the Orkney volcanics is a fairly complex one. Therefore, in order to give the reader a fair chance in judging the results it has been necessary to present a relatively large number of examples of the directional behaviour with increasing temperature. Thus, figs. 3 and 4 give altogether the directional results of 24 specimens from 16 different samples.

Below an average remanence intensity of about 5.10^{-6} emu/cm^3 for lava specimens and about 5.10^{-7} emu/cm^3 for the ashy sediment the measured magnetization vector initiates an irregular behaviour, i.e. a well-defined directional trend associated with a smooth variation of the corresponding remanence intensity with temperature is superseded by scattered directional results as well as irregular intensity variations. This latter stage has obviously no geophysical relevance and the associated directions have therefore not been included in figs. 3 and 4. The critical intensity level (where scattered results occur) of the strongly oxidized ashy sediment is very well defined while that of the lavas varies between 1.10^{-5} emu/cm^3 and 3.10^{-6} emu/cm^3.

In the majority of lava specimens the direction of NRM has a steep positive (downward) inclination. On demagnetization to temperatures below approximately 550°C the general behaviour is a progressive directional movement (probably resulting from the elimination of a viscous component) towards a southerly direction with low positive inclination. There are, however, very few cases in which this latter direction (carried by a spinel phase) can be satisfactorily determined. In some cases the reason for this is that the above mentioned random scatter in the results becomes significant before a stable end point can be verified. Furthermore, haematite (which has a high thermal stability) contributes in varying degree to the NRM of these lavas. This latter component is in general directed opposite to the high temperature direction of the spinel phase. Therefore, at temperatures in excess of approximately 500°C a northerly directional movement may commence before the direction of the reversed component can be established. Thus, it is only in a few cases where the haematite fraction contributes a very small proportion of the NRM intensity that the high temperature direction of the spinel phase (for example, the mean of the encircled directions of specimens OL3-D 1 and OL16-A2) can be experimentally determined with reasonable certainty.

In a few lava specimens there are no certain indications that a spinel component is present as a re-

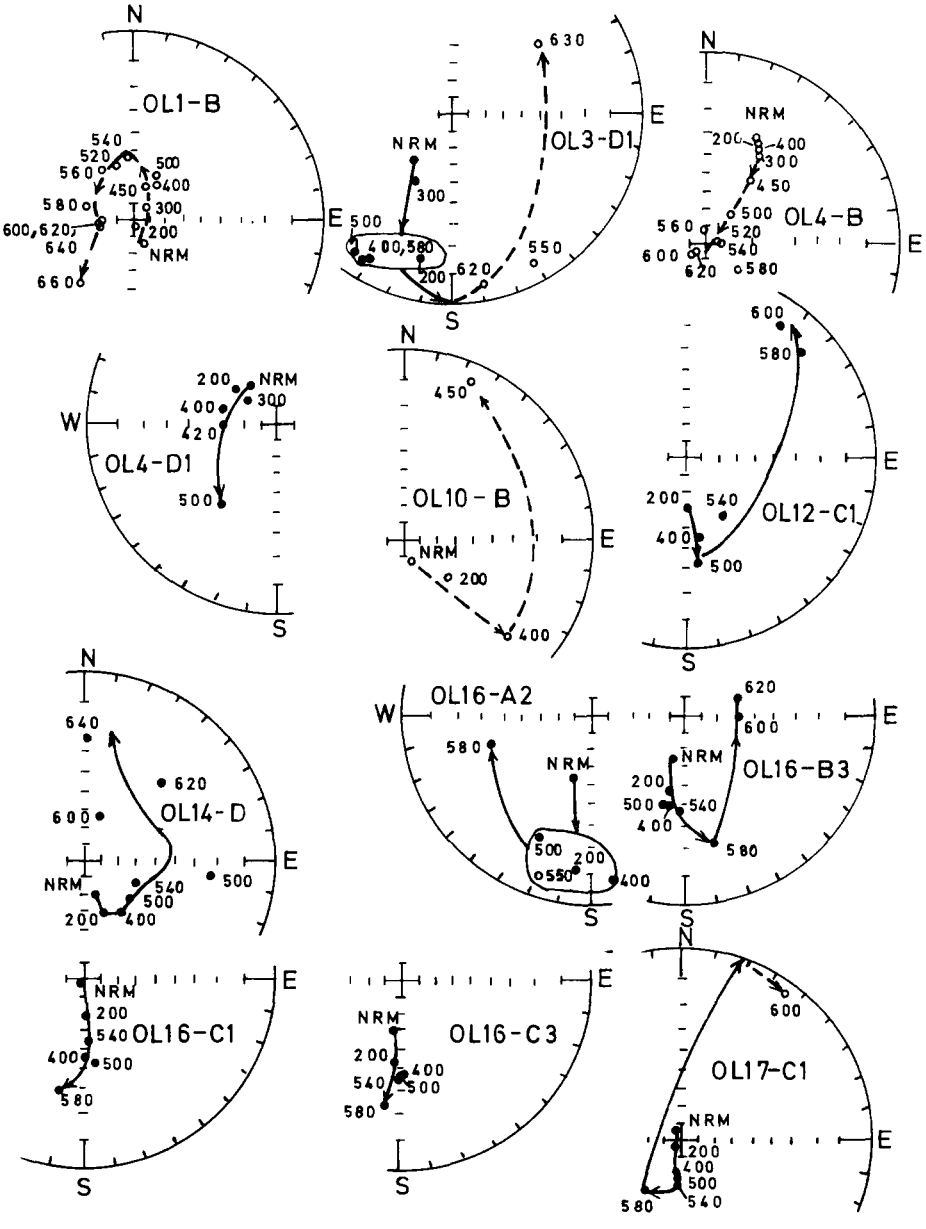


Fig. 3. Direction of remanent magnetization of lava specimens as a function of temperature.

manence carrier, as the thermal decay curves are nearly linearly decreasing up to temperatures close to the T_c of haematite. Demagnetization to high temperatures may reveal a haematite component directed southerly (cf. specimens OL1-B and OL4-B of fig. 3). The remanent magnetization of these latter specimens may well consist of two haematite components of opposite polarity.

Also the NRM of the ash deposit consists of two oppositely directed components. These directions are both associated with haematite but in general the reversed component exhibits somewhat higher blocking temperatures than the normal component. As the relative abundance of these two components varies, scattered intermediate directions of fairly high stabil-

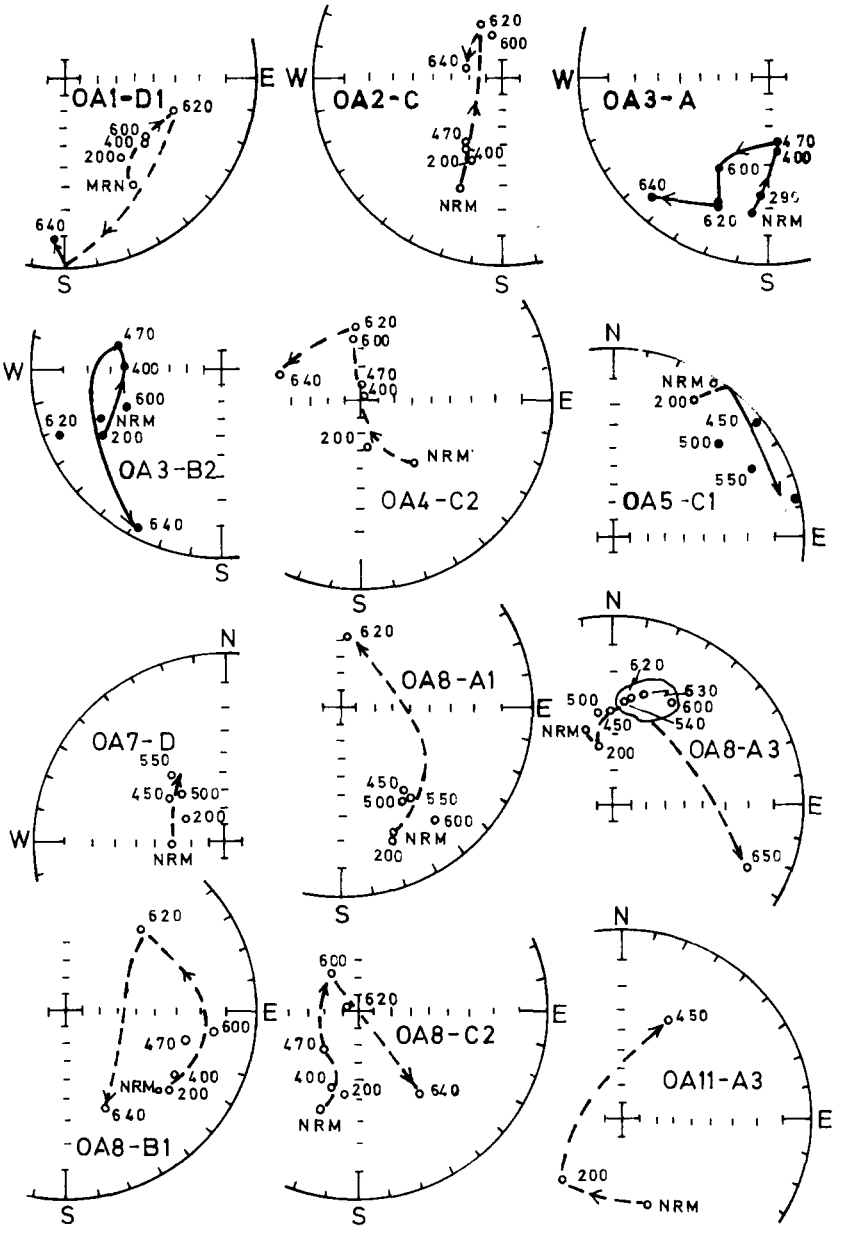


Fig. 4. Direction of remanent magnetization of ashy specimens as a function of temperature.

ity may result. The experimental problem of estimating the high stability direction is exactly as encountered in previous studies on highly oxidized rocks [9, 14, 16], i.e. a single high temperature direction has to be sought in a very narrow temperature range just below the T_c of haematite (~ 670°C). Owing to this difficulty an acceptable estimate of the high stability component is very rare. However, it appears fairly certain that the directional trends of most of the ashy specimens tend to converge in the direction

Table 1
Summary of palaeomagnetic properties of European Old Red Sandstone rocks.

Ref.	Formation	Age	Pole	Polarity	Remarks
[16]	Ringerike sed. S. Norway	Su	159E, 21N	N	The formation became nearly completely remagnetized during Permian igneous activity in the area. The polarity is that of the stable (high temperature) haematite component.
[9]	Cheviot lavas S. Scotland	D1		N	These lavas are strongly oxidized and exhibit a composite magnetization. Single remanence components have not been estimated with sufficient accuracy. The polarity is that associated with the high temperature haematite component.
[1]	Anglo-Welsh Cuvette sed. S. England	D1		N	The formation became nearly completely remagnetized in Permo-Carboniferous times. In the opinion of the present authors the stable haematite component has not been satisfactorily determined. The polarity is that displayed by the high temperature (haematite) component.
[17]	Røragen sed. S. Norway	D1-m	160E, 19N	M	
[18]	Kvamshesten sed. S. Norway	Dm	170E, 22N	R	Boundary to underlying formation constitutes a thrustplane. Therefore, the deviation from the other Norwegian ORS results may be explained by a small anti-clockwise rotation of the Kvamshesten Old Red.
[20]	Lamprophyre dike Central Norway	Dm-u (363 ± 15 my)		R	The dike is less than one metre wide. Magnetic self-reversal properties occur during heating experiments. The polarity is that of specimens not exhibiting the self-reversal property.
This paper	Orkney volcanics	Dm-u	149.5E, 23.7N	R	Pole position considered somewhat inaccurate owing to the problems of estimating single remanence components.

of the stable spinel component of the lavas (cf. for instance 0A1-D1, 0A3-A and 0A3-B2 of fig. 4).

Because of a lower thermal stability the normal component of the ash deposit is even more difficult to determine than the reversed one. In some cases, however, where the normal magnetization predominates in the total remanence (cf. for instance 0A5-C1 and 0A8-A3 of fig. 4) a direction 180° opposite to the reversed direction of these specimens is suggested. Also, such a direction would agree with the haematite direction of normal polarity revealed by many lava specimens.

5. Origin of the remanent magnetization

The high oxidation state of the ore minerals concerned together with the fact that haematite contributes two remanence directions of opposite polarity suggests strongly that chemical magnetization (CRM) plays an important role in the Orkney volcanics. This assumption is further strengthened by the predominance of low temperature oxidation features. The experimental results indicate that the time span of chemical changes involves one or more field inversions, but there is no evidence of relative polar migration.

Contrary to the remanence of the haematite, the high temperature direction of the spinel phase appears to be exclusively of one polarity (reverse). This indicates that the magnetization of the spinel phase is more restricted in time than the magnetization carried by haematite. With reference to our previous reasoning (sect. 3) it seems likely, therefore, that the spinel component reflects the geomagnetic field at about the time the igneous activity took place.

This latter magnetization also fits excellently into the polarity/time sequence as obtained for some Old Red Sandstone (ORS) rocks of Norway and Great Britain [7]. These results, which are listed in table 1, indicate a polarity change from a normal sense in early ORS time to a reversed one in late ORS time. The transition period is defined by the Rødragen red beds of Southern Norway which are of late Lower-Middle Devonian age. Despite the high oxidation state in most of these cases, rendering a time-fixing of the fossil magnetization rather suspect, the observed trend in the polarity as a function of rock age indicates that the palaeomagnetic directions concerned are not essentially younger than the origin of the rocks.

It has been suggested that the pole of rotation relative to Europe remained fixed throughout a major part of the Palaeozoic, the only polar shift reasonably documented so far occurring around Middle Carboniferous [22]. Therefore, information about the geomagnetic polarity in the Upper Devonian and in the Lower Carboniferous would be of essential importance for the interpretation of the magnetization build-up in the Orkney volcanics. Unfortunately, at present there appears to be no reliable data for the uppermost Devonian. On the other hand, there is evidence that both polarities have existed in the Lower Carboniferous field [19]. This would imply that chemical magnetizations covering a time span of about 50 my (Middle Devonian-Middle Carboniferous) may be present in the rocks considered.

Suggesting that the high temperature spinel phase direction (reverse) of the Orkney lavas does not postdate the igneous activity significantly (at least representing the relevant polarity epoch), the haematite direction of normal polarity must be of later origin. This latter component has probably been imposed either in the uppermost Devonian or in the Lower Carboniferous. Furthermore, it is possible that the reversed haematite component, which is of particular importance in the ash layer, does not postdate the volcanic activity significantly. The original loose character of the ash deposit probably enabled the low temperature oxidation processes to proceed very rapidly. The reversed haematite direction which appears to exist in some strongly oxidized lava specimens may well be of deuteric origin in part.

The consequence of opposite-direction components within single specimens is in general the occurrence of deviating remanence directions. The relative abundance of normal and reversed components may vary greatly, resulting in scattered but fairly stable remanence directions even within single hand samples. An example is shown by specimens of sample 0A8 in fig. 4. A more elaborate discussion of the palaeomagnetic problems encountered here is given elsewhere [8].

With reference to the foregoing section, there are very few specimens in which one of the existing magnetization components can be estimated with reasonable reliability. A necessary requirement for acceptance of palaeomagnetic data is that there must be experimental verification that systematic direction changes have come to an end. However, if such a stable direction is associated with the spinel phase (as in 0L3-D1 and 0L16-A2 of fig. 3) any oppositely-directed component existing above $\sim 580°C$ must be of negligible importance. From a total of about 60 tested specimens, only 7 high temperature directions (2 normal and 5 reversed) satisfy these requirements. The reliability of the estimated magnetization axis is further strengthened by the large number of directional paths tending to converge towards the two antiparallel groups obtained. The data based on the acceptable directions are given in table 2.

In a case of extensive low temperature oxidation, the traditional concept that a specimen direction represents a spot reading of an ancient field is unlikely to apply. In such cases it seems more realistic to consider a specimen direction to represent the mean magnetization acquired over a somewhat longer time interval. Thus, the time span incorporated in a palaeomagnetic direction of a specimen is probably much longer than the period of geomagnetic secular variation. This means that the scatter of specimen directions should be largely due to causes other than field changes. In the present case it appears likely that the main scatter

Table 2
Palaeomagnetic results from the Orkney volcanics.

N	R	k	α_{95}	Dm	Im	Pole	Remarks
7	6.9	44.8	9.1	205	8.4	23.7N, 149.5E	Unit weight on specimens. Statistical parameters are according to Fisher [21].

is associated with the experimental problems of establishing single remanence components. However, despite the small amount of acceptable data from the Orkney volcanics, rendering the palaeomagnetic result somewhat uncertain, the estimated pole position is in fairly good agreement with those based on Norwegian ORS rocks [7, 16–18], i.e. the Orkney result provides further evidence that the ORS pole relative to Europe is situated at approximately 160°E, 20°N in the present grid system (cf. table 1).

6. Conclusion

The Orkney volcanics provide another example of the complexity encountered in the palaeomagnetic record. A composite magnetization even within a single lava specimen (a striking contrast to the original magnetic structure) has recently also been demonstrated in some Permian lava flows of Southern Norway [14] and there appears to be a large number of potential examples in the literature [8]. Thus, the most recent palaeomagnetic data from the Lower Devonian lavas of the Caledonian basin of Scotland [4] show strong indications of possessing a composite build-up in that the normal as well as the reversed directions (cleaned for low stability components) exhibit a marked meridional distribution. Therefore, anticipating the remagnetization problems of this latter formation, there are strong reasons for believing that the data from the Caledonian basin of Scotland presently available do not reflect the true Devonian geomagnetic field. The Devonian polarity sequence defined so far by 7 different rock formations may be an observation of exceedingly great importance in future investigations; besides its importance for geomagnetism, one of the obvious applications is as a tool in Old Red Sandstone correlation.

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References

[1] F.H. Chamalaun, Origin of secondary magnetization of the Old Red Sandstone of the Anglo-Welsh Cuvette, J. Geophys. Res. 69 (1964) 4327.
[2] B.J.J. Embleton, Laboratory stability tests applied to Devonian lavas from Scotland, Geophys. J. 16 (1968) 239.
[3] B.J.J. Embleton, Some recent Devonian palaeomagnetic data for Europe, in: Palaeogeophysics, ed. S.K. Runcorn (Academic Press, London, 1970).
[4] E.W. McMurry, Palaeomagnetic results from Scottish lavas of Lower Devonian age, in: Palaeogeophysics, ed. S.K. Runcorn (Academic Press, London, 1970).
[5] K.M. Storetvedt, A discussion of the Devonian pole for Europe, Tectonophysics 4 (1967) 155.
[6] K.M. Storetvedt, On remagnetization problems in palaeomagnetism, Earth Planet. Sci. Letters 4 (1968) 107.
[7] K.M. Storetvedt, The Devonian palaeomagnetic field for Europe, in: Palaeogeophysics, ed. S.K. Runcorn, (Academic Press, London, 1970).
[8] K.M. Storetvedt, On remagnetization problems in palaeomagnetism; further considerations, Earth Planet. Sci. Letters 9 (1970) 407.
[9] K.M. Storetvedt and E. Halvorsen, On the palaeomagnetic reliability of the Scottish Devonian lavas, Tectonophysics 5 (1968) 447.
[10] G.V. Wilson, W. Edwards, J. Knox, R.C.B. Jones and J.V. Stephens, The geology of the Orkneys, H.M. Stationery Office, Edinburgh (1935).
[11] J. Verhoogen, Oxidation of iron-titanium oxides in igneous rocks, J. Geol. 70 (1962) 161.
[12] P. Readman and W. O'Reilly, Synthesis and inversion of non-stoichiometric titanomagnetites, Phys. Earth Planet. Int. 4 (1970) 121.
[13] J.M. Ade-Hall, M.A. Kahn, P. Dagley and R.L. Wilson,

A detailed opaque petrological and magnetic investigation of a single Tertiary lava flow from Skye, Scotland – I. Iron-titanium oxide petrology, Geophys. J. 16 (1968) 375.

[14] K.M. Storetvedt and N. Petersen, On chemical magnetization in some Permian lava flows of Southern Norway, Z. Geophys. 36 (1970) 569.

[15] H. Soffel, Die Bereichsstrukturen der Titanomagnetite in zwei tertiären Basalten und die Beziehung zu makroskopisch gemessenen magnetischen Eigenschaften dieser Gesteine, Thesis, Univ. of Munich (1968).

[16] K.M. Storetvedt, E. Halvorsen and G. Gjellestad, Thermal analysis of the natural remanent magnetism of some Upper Silurian Red Sandstones in the Oslo region, Tectonophysics 5 (1968) 413.

[17] K.M. Storetvedt and G. Gjellestad, Palaeomagnetic investigation of an Old Red Sandstone formation of Southern Norway, Nature 212 (1966) 59.

[18] L.G. Lie, K.M. Storetvedt and G. Gjellestad, The palaeomagnetism of the Kvamshesten Old Red Sequence, Southwest Norway, Norsk Geol. Tidsskr. 49 (1969) 241.

[19] R.L. Wilson and C.W.F. Everitt, Thermal demagnetization of some Carboniferous lavas for palaeomagnetic purposes, Geophys. J. 8 (1963) 149.

[20] K.M. Storetvedt, Magnetic properties of an ultrabasic biotite lamprophyre dike from the island of Ytterøy, Norway, Norsk Geol. Tidsskr. 47 (1967) 171.

[21] R.A. Fisher, Dispersion on a sphere, Proc. Roy. Soc. London, Ser. A, 217 (1953) 295.

[22] K.M. Storetvedt, A synthesis of the Palaeozoic palaeomagnetic data for Europe, Earth Planet. Sci. Letters 3 (1968) 444.